

Jasim Kareem Badawi

SENIOR AI & SOFTWARE DEVELOPER | SECURITY RESEARCHER

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PROFESSIONAL SUMMARY

Highly technical, self-taught Software Architect and Security Researcher with hands-on experience designing and deploying end-to-end distributed systems, security-hardened fintech interfaces, and agentic workflows. Proven track record of building production-grade tools, including financial firewalls for AI models, visual messaging automation engines with local banking integrations, and neobanking API architectures for autonomous agents. Combines deep technical execution (Python, Node.js, Go, Cloudflare Workers, Docker) with a strong analytical focus to design highly intuitive, ethical digital products.

TECHNICAL SKILLS

AI & AGENTIC TECHNOLOGIES:	AI Agents, LLM Fine-tuning & Deployment, Retrieval-Augmented Generation (RAG) Architectures, Prompt Engineering & Symbolic Prompts
FINTECH & SECURITY:	Atomic Transactions, HMAC SHA-256 Signature Verification, AES-256-GCM Encryption, JWT Authentication, API Key Hashing (SHA-256), RLS Policies (Supabase)
BACKEND & SYSTEM DESIGN:	Node.js, Express, Go, Python, FastAPI, Cloudflare Workers, Hono, RESTful APIs, WebSockets, Redis, BullMQ, Docker (Multi-container environments)
FRONTEND & DATABASES:	React, TypeScript, Tailwind CSS, Vite, HTML5/CSS3, shadcn/ui, PostgreSQL, Supabase, Prisma ORM, Database Pagination & Optimization

FEATURED PROJECTS

Agmoney: Financial Guardian & Proxy for AI Agents agmoney.aiandthings.tech | [GitHub](#)

Architected and deployed a "Financial Firewall" proxy sitting between autonomous AI agents and LLM APIs to enforce hard budget caps in real-time.

- Developed a low-latency gateway using Cloudflare Workers and Hono with full real-time chunk streaming support.
- Secured OpenAI Master Keys on the client side using AES-256-GCM encryption with PBKDF2 key derivation.
- Engineered atomic cost calculation and database syncing using `ctx.waitUntil()` on Cloudflare Workers and Supabase (PostgreSQL).

SumerSend: Enterprise Messaging & Secured Payments Engine sumersend.aiandthings.tech | [GitHub](#)

Developed an enterprise-grade visual messaging builder featuring direct email MX delivery, custom domain DNS validation (SPF/DKIM), and a secure billing/top-up module.

- Integrated local payment gateways (Zain Cash, AsiaHawala, FastPay, Qi Card) verifying webhook validity using HMAC SHA-256 signatures.
- Designed and executed a comprehensive security remediation suite resolving RLS bypasses, JWT session leakages, Stored XSS vulnerabilities, and double-spend attacks via atomic database transactions (`topup_wallet_atomic`).
- Built a campaign distribution module running off-thread via Redis and BullMQ.

Bank of Agents (مصرف الوكلاء): API-First Neobank for AI Agents bank.aiandthings.tech | [GitHub](#)

Built a prototype neobanking platform designed specifically for the machine economy, enabling autonomous agents to hold digital wallets, check balances, and spend within predefined rules.

- Implemented an agent-centric API verified through distributed secure agent keys.
- Programmed a human-in-the-loop (HITL) transaction approval mechanism, enabling users to approve or reject pending agent payments from a responsive dashboard.
- Structured database relations using Prisma ORM with PostgreSQL.

SECURITY RESEARCH & VULNERABILITY AUDITING

Independent Security Researcher JUNE 2025 – PRESENT

- Facebook Privacy Bypass Research:** Discovered and documented a privacy vulnerability on the Facebook platform, authoring a full Proof of Concept (PoC) and technical report outlining mitigation steps.
- OTP Protocol Vulnerability Analysis (OTPiQ):** Developed custom python fuzzers and protocol testing scripts to analyze and verify weaknesses in regional OTP (One-Time Password) verification endpoints.
- Vulnerability Tooling Development:** Built custom lightweight penetration testing tools in Python (`'wp_advanced_scanner.py'`, `'wp_xss_deep_verify.py'`) to automate the detection of SQL injections, cross-site scripting (XSS), and logic flaws in web applications.

PROFESSIONAL EXPERIENCE

Independent Software Developer & Freelancer | RemoteJULY 2023 – PRESENT

- Designed, built, and launched complex end-to-end applications (Agmoney, SumerSend) to master modern system design, cryptography, and serverless architectures.
- Developed custom web platforms, automated workflows, and secure API integrations for remote clients.
- Contributed to open-source tools, focusing on backend security optimization, database performance, and developer-focused technical specifications.

Project Manager | Private Family LibrarySEPTEMBER 2024 – OCTOBER 2025

- Managed daily business operations, inventory, and procurement cycles.
- Supervised customer relations and team coordination to achieve consistent targets.
- Built and maintained a digital system to track and audit financial transactions.

EDUCATION

Bachelor of Arts (BA) in English Language | University of BasrahGRADUATED: JULY 2023

Self-Taught Computer Science & Security Engineering | Independent StudyONGOING